FUNDAMENTALS OF URBAN AND REGIONAL PLANNING

ARCH-305 : Arch & Town Planning



Contents

- Course contents/outline
- Introduction to Town Planning
- Aims of Town Planning
- Principles of Town Planning
- Level of Planning
- History of Planning

Overview of Planning

- Planning as one of the basic “social driver” of society.
- It is treated as both science and art.
- Planning Age
- There are many forms of planning.
- Certain features of planning is common like sequence of actions, planning problems etc.

Terminologies and Definitions:

What is Planning ? (1)

- Abundant individual definition, yet no general theory of planning to clearly identify what constitute planning.

What is Planning?

- Definition of planning is subjective.
- Planning is pre- thinking and pre-arranging things.

What is Planning ?(2)

Planning is a

future-oriented

problem solving

process.







Definition of Planning

(Friedman, An Introduction to Regional Planning, J. Glasson)

"Planning is primarily a way of thinking about social and economic problems, planning is oriented predominantly towards future, is deeply concerned with the relation of goals to collective decision and strive for comprehensiveness in policy and program."

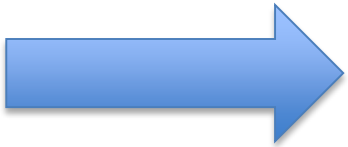
Town Planning (1)

The art and science of ordering the use of land and siting of buildings and communication routes so as to secure the maximum practicable degree of economy, convenience and beauty within available resources and time frame

(Lewis Keeble).

Town Planning (2)

- Friedmann defines planning as the component that links knowledge & action.

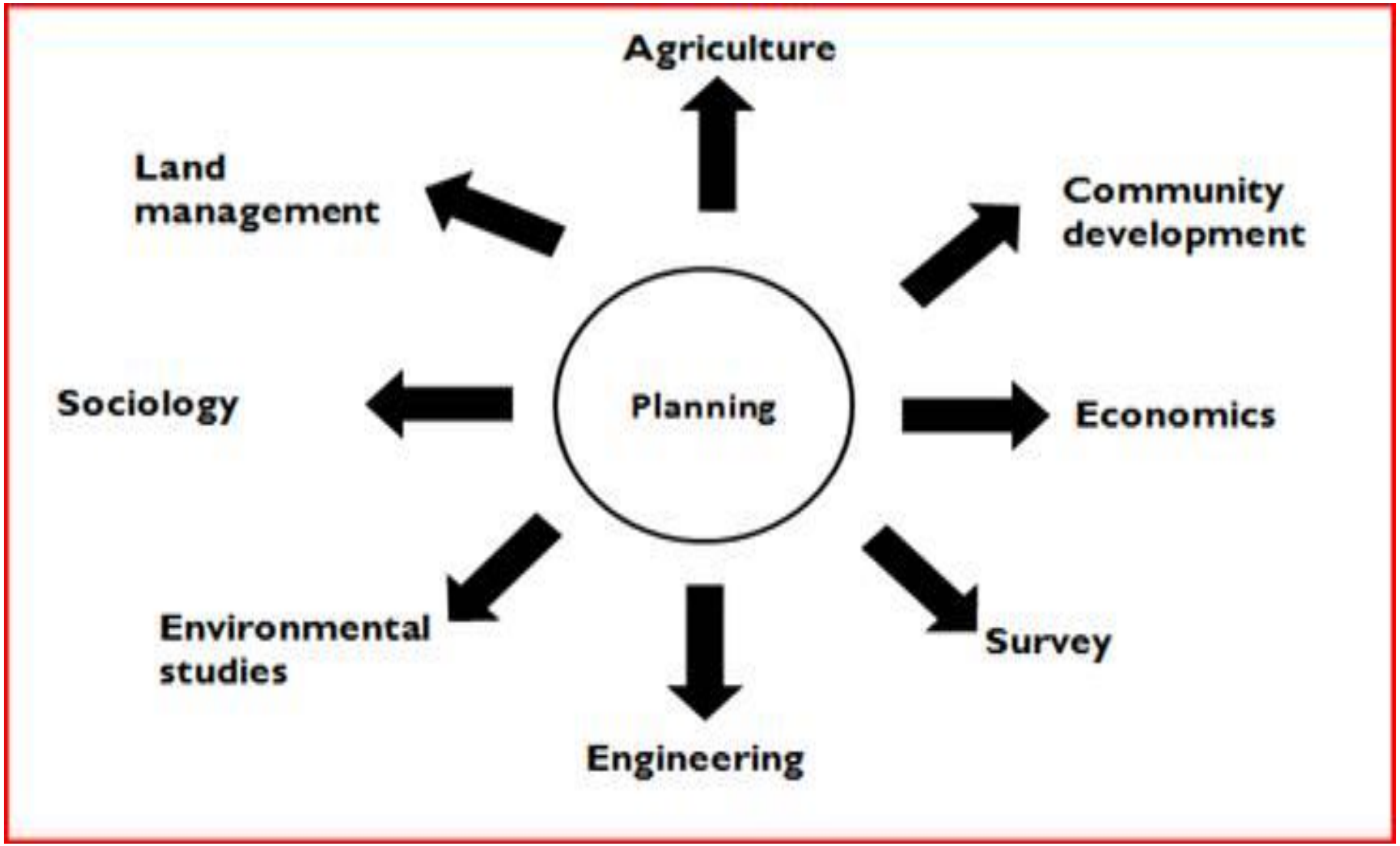
KNOWLEDGE  ACTION

- Planning is what links technical knowledge and actions in the **public domain**.

Similar terminologies

- Urban Planning
- City Planning
- Infrastructure Planning
- Urban and Regional Planning
- Spatial Planning and Development
- Urban Planning and Development

Planning is Interdisciplinary



Urban Planning - Urban Design - Architecture Axis

- The relationship is historical:
- In the 1960s, planning and architecture were split
  - Planning concentrated on land use patterns and socio-economic issues (macro)
  - Architecture concentrated on the design of buildings (micro)

- There emerged a responsibility gap where design of public space was concerned
- Urban design came in to bridge this gap

Town/City?

- A Town or a city is a place of urban living.
- Urban means an environment in which natural surroundings are dominated by the man-made surroundings.
- Economic base, infrastructure and population density are the major criteria.
- Cities are engines of economic growth and epitomes of civilization.

Town planning is considered as an art and science of shaping and guiding the physical growth of the town to meet the needs of its population.

Defining City

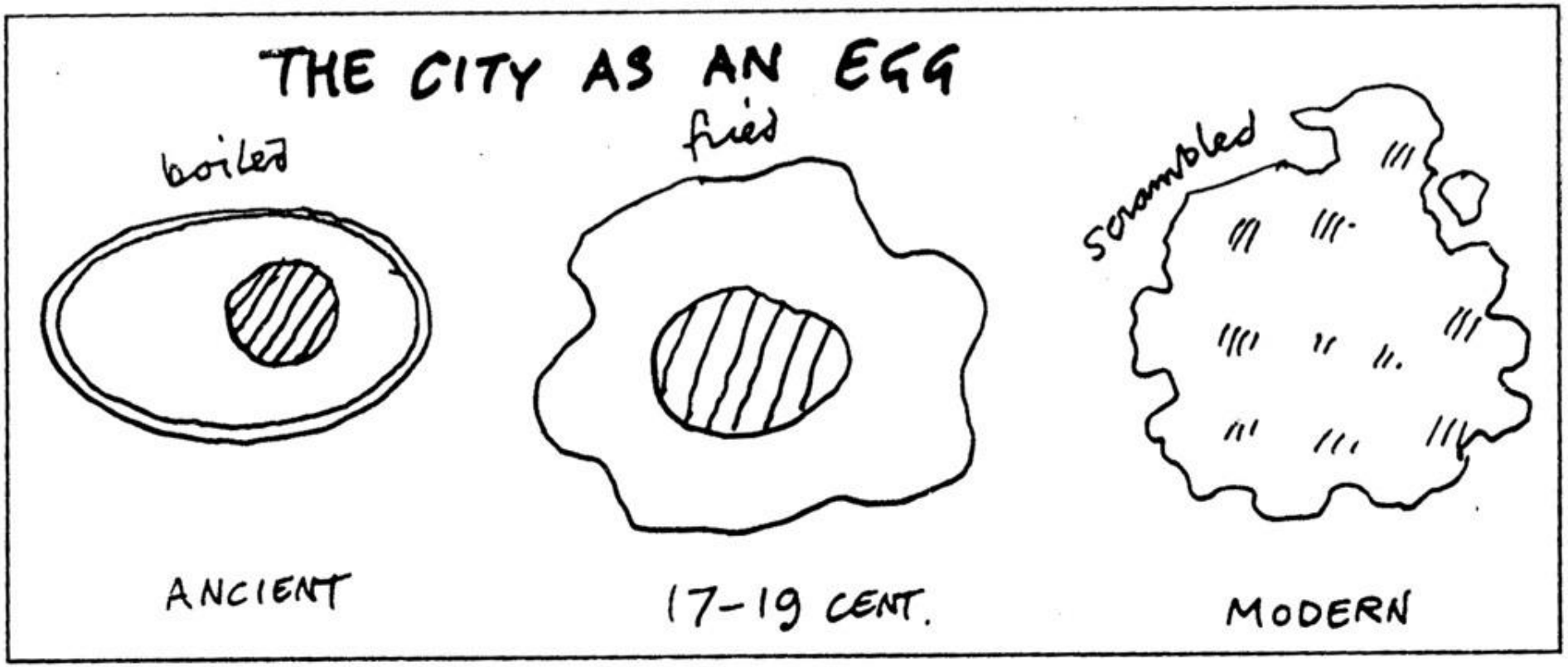
- The term *city* may be defined as a relatively large, dense, and permanent settlement of people engaged in diverse economic activities.
- A more technically oriented definition of the term city is a complex arrangement of land uses, linked together by circulation systems and made viable by other systems of utilities such as sewerage, water supply, electric power, and telecommunications.

Characteristics of City

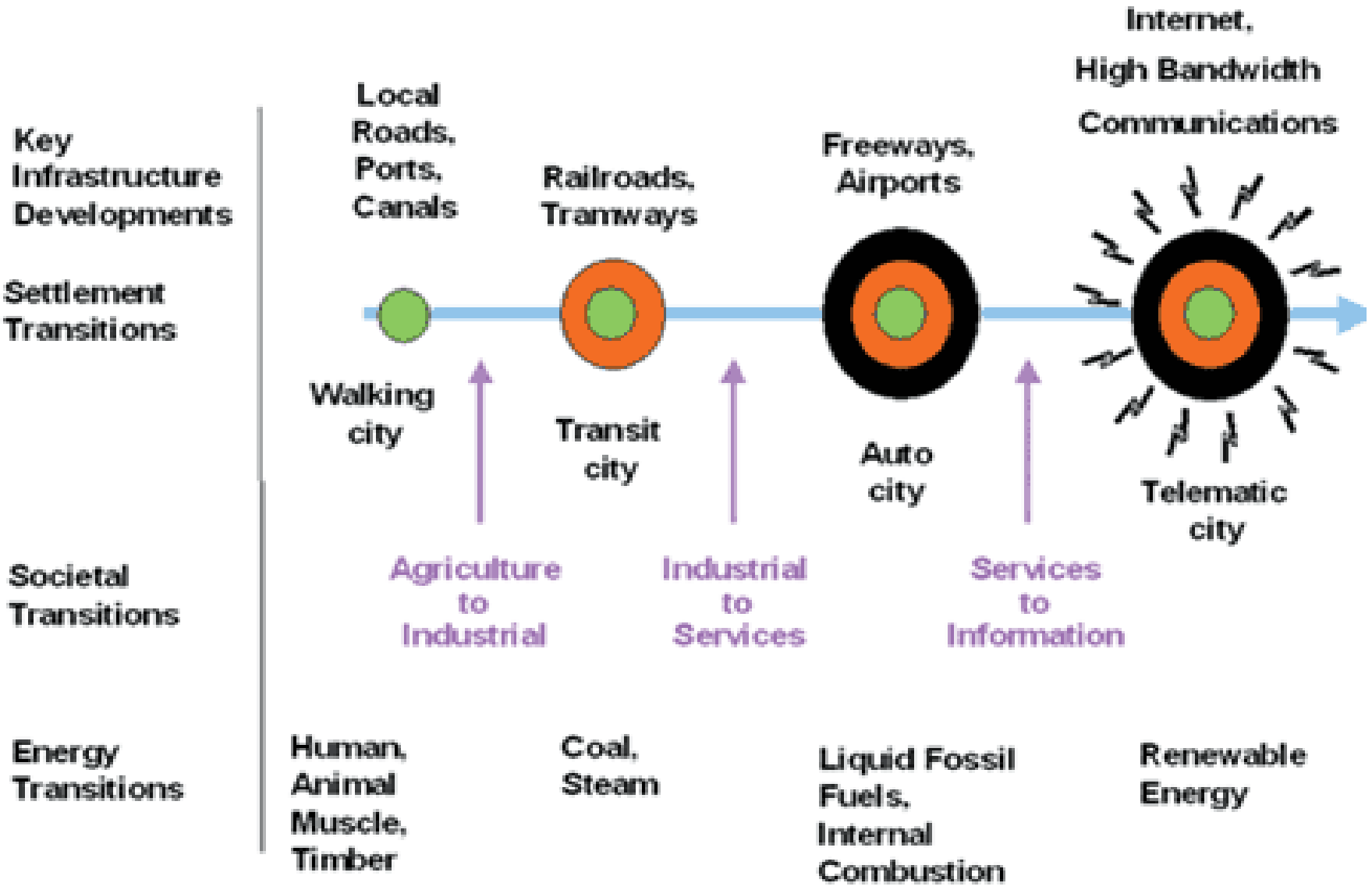
Elements of a city

- Transportation routes
- Residential (Neighborhood)
- Commercial (Central Business District)
- Public building
- Recreational facilities (park)
- Industries
- Urban agriculture
- Power stations etc

Evolution of Cities



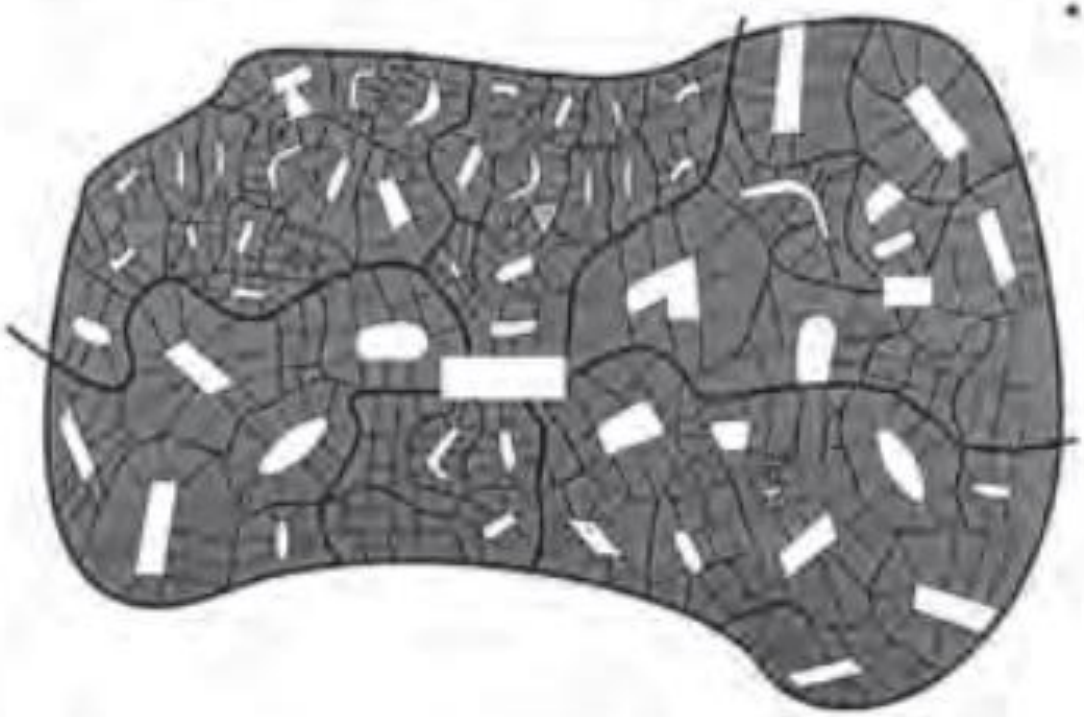
Evolution of Cities

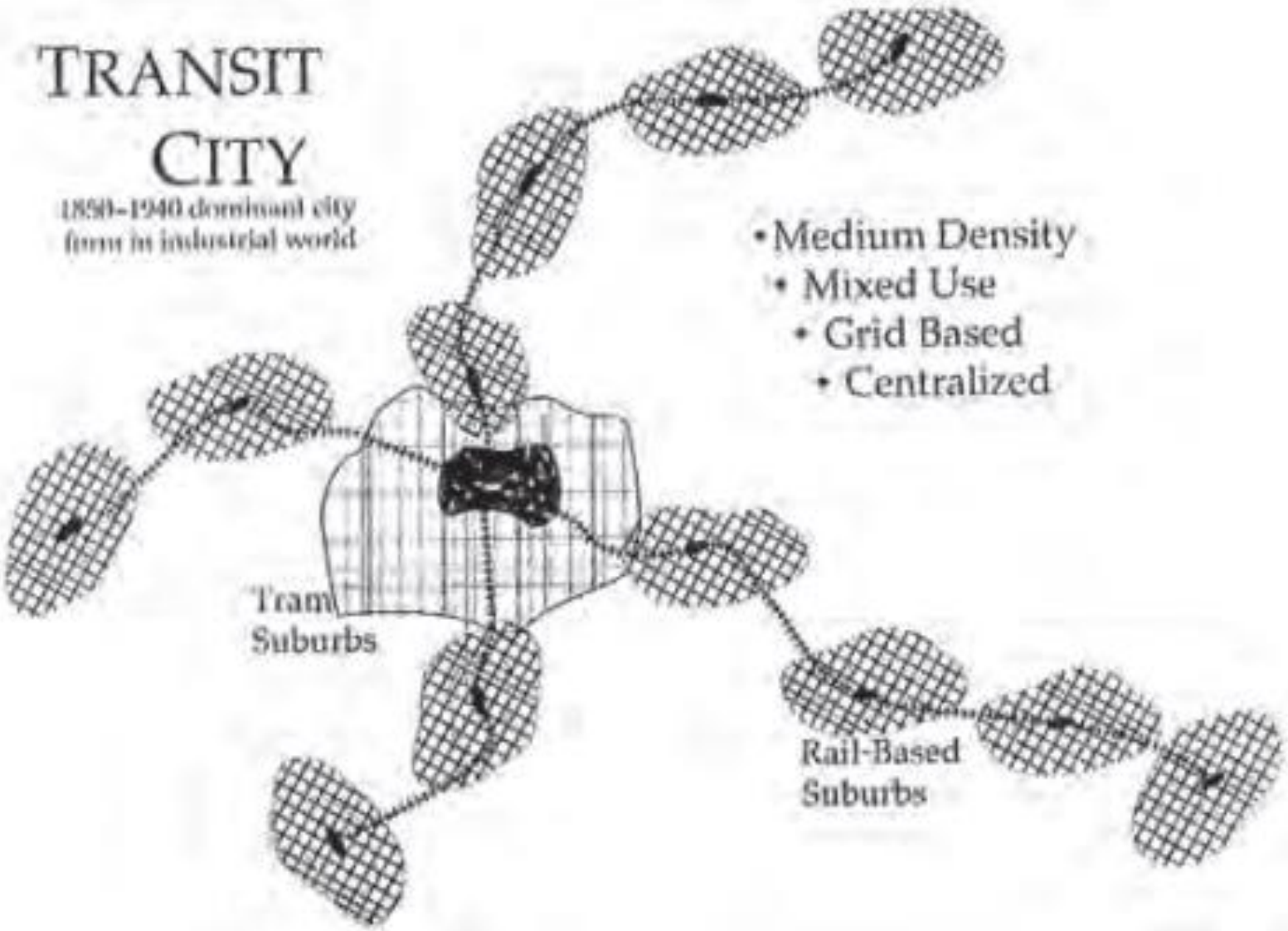


TRADITIONAL WALKING CITY

Up to 1850 in Europe

- High Density
- Mixed Use
- Organic Structure

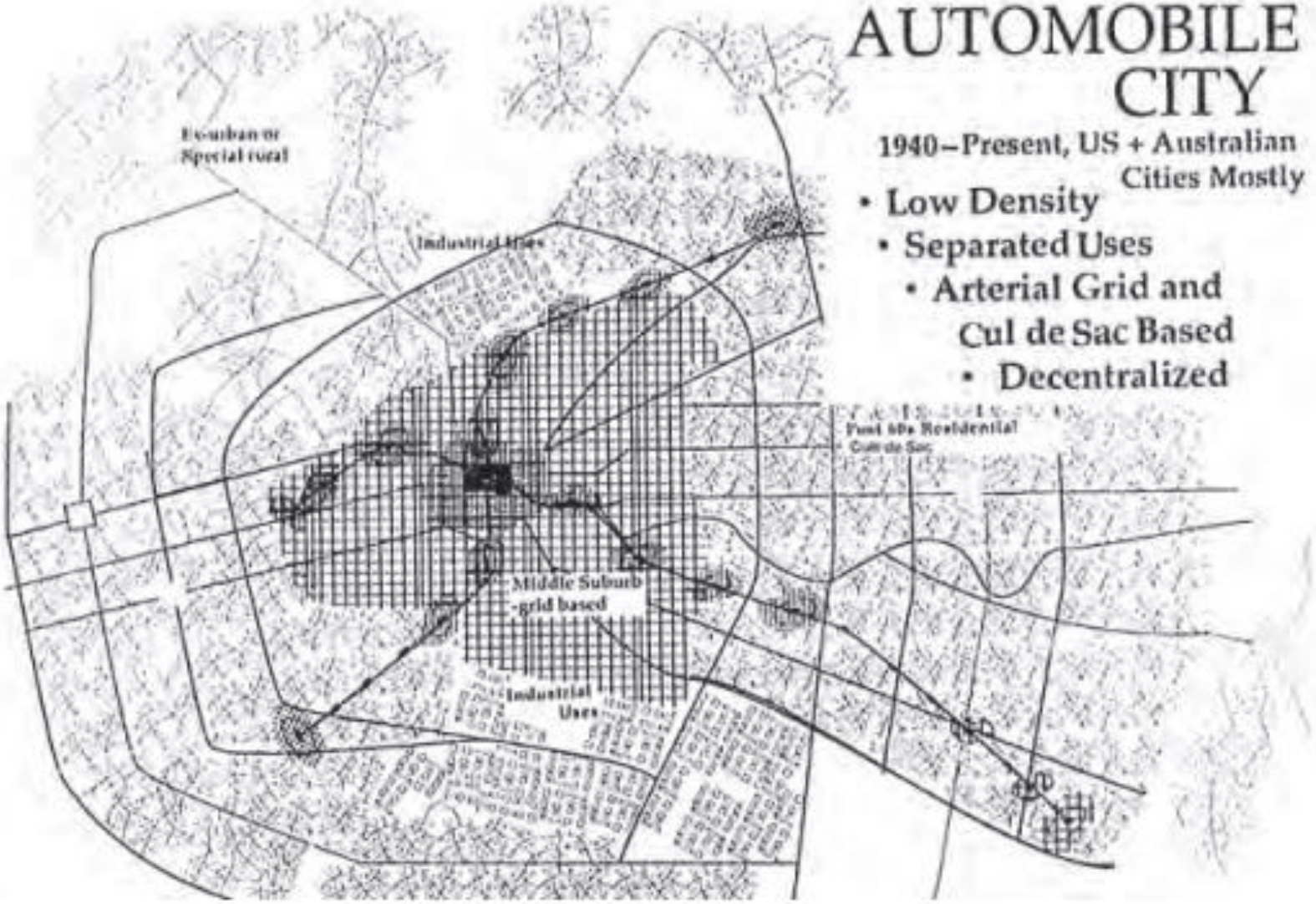



TRANSIT CITY
1850–1940 dominant city form in industrial world
• Medium Density
• Mixed Use
• Grid Based
• Centralized
Tram Suburbs
Rail-Based Suburbs

AUTOMOBILE CITY

1940–Present, US + Australian Cities Mostly

- Low Density
- Separated Uses
- Arterial Grid and Cul de Sac Based
- Decentralized



City as an Economic Entity

- Engines of Economic Growth
  - ***Commerce and Trade*** (Karachi)
  - ***Industry and Manufacturing*** (Faisalabad)
  - ***Government and Administration*** (Islamabad)
  - ***Mixed Functionalities*** (Lahore)
  - ***Recreation and Health*** (Murree)

Ekistics:

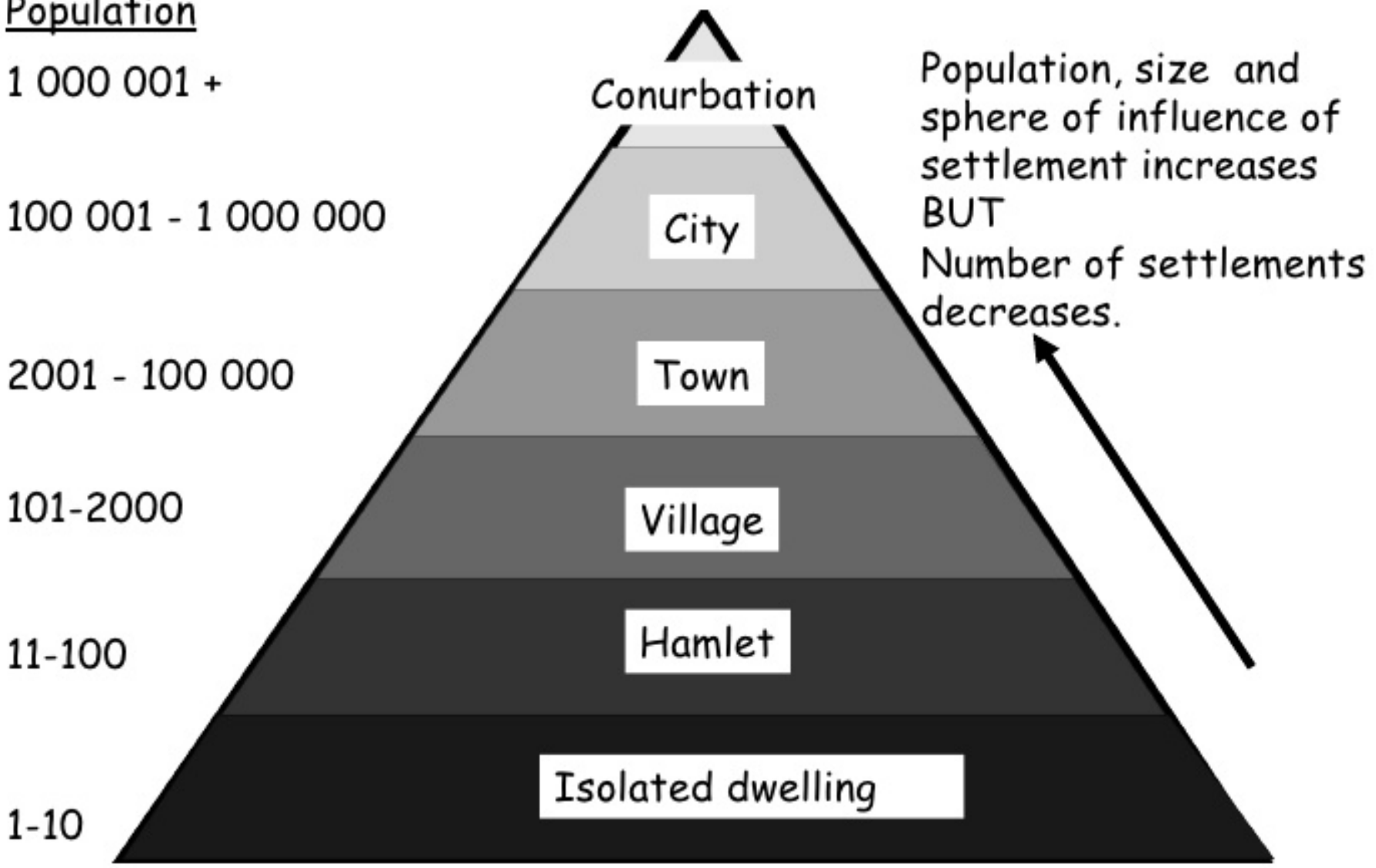
The Science of Human Settlements

EKISTICS

Ekistics was introduced as the science of human settlements (Doxiadis, 1968).

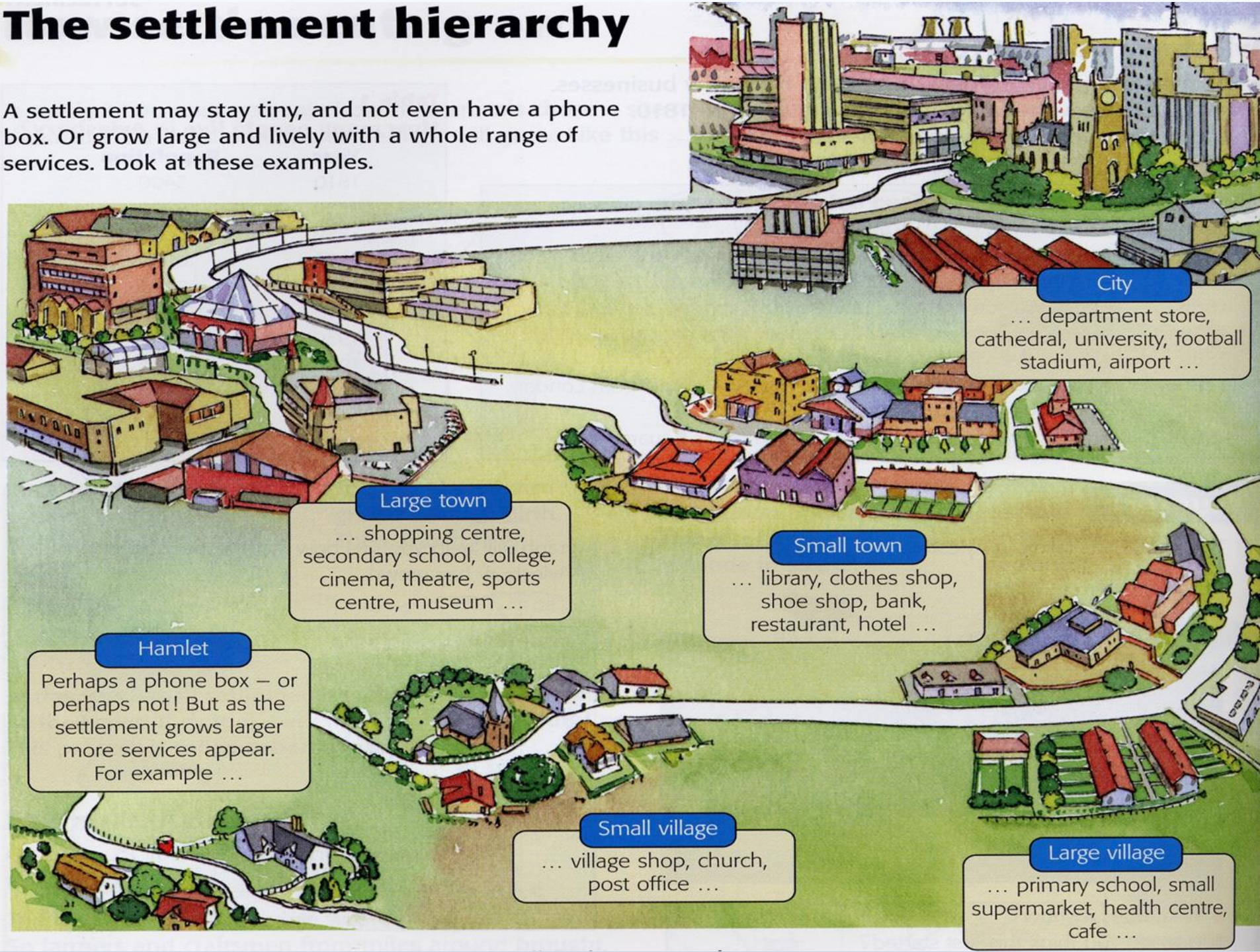
- **Minor shells** or elementary units- **man (anthropos), room, house;**
- **Micro-settlements**- units smaller than, or as small as, the traditional town where people used , do & still do achieve interconnection by walking **(housegroup, small neighbourhood);**
- **Meso-settlements**- between traditional town & conurbation within which one can commute daily **(small polis, polis, small metropolis, small megalopolis, small eperopolis, eperopolis);** and
- **Macro-settlements**- whose largest possible expression is the **Ecumenopolis.**

The Settlement Hierarchy



The settlement hierarchy

A settlement may stay tiny, and not even have a phone box. Or grow large and lively with a whole range of services. Look at these examples.

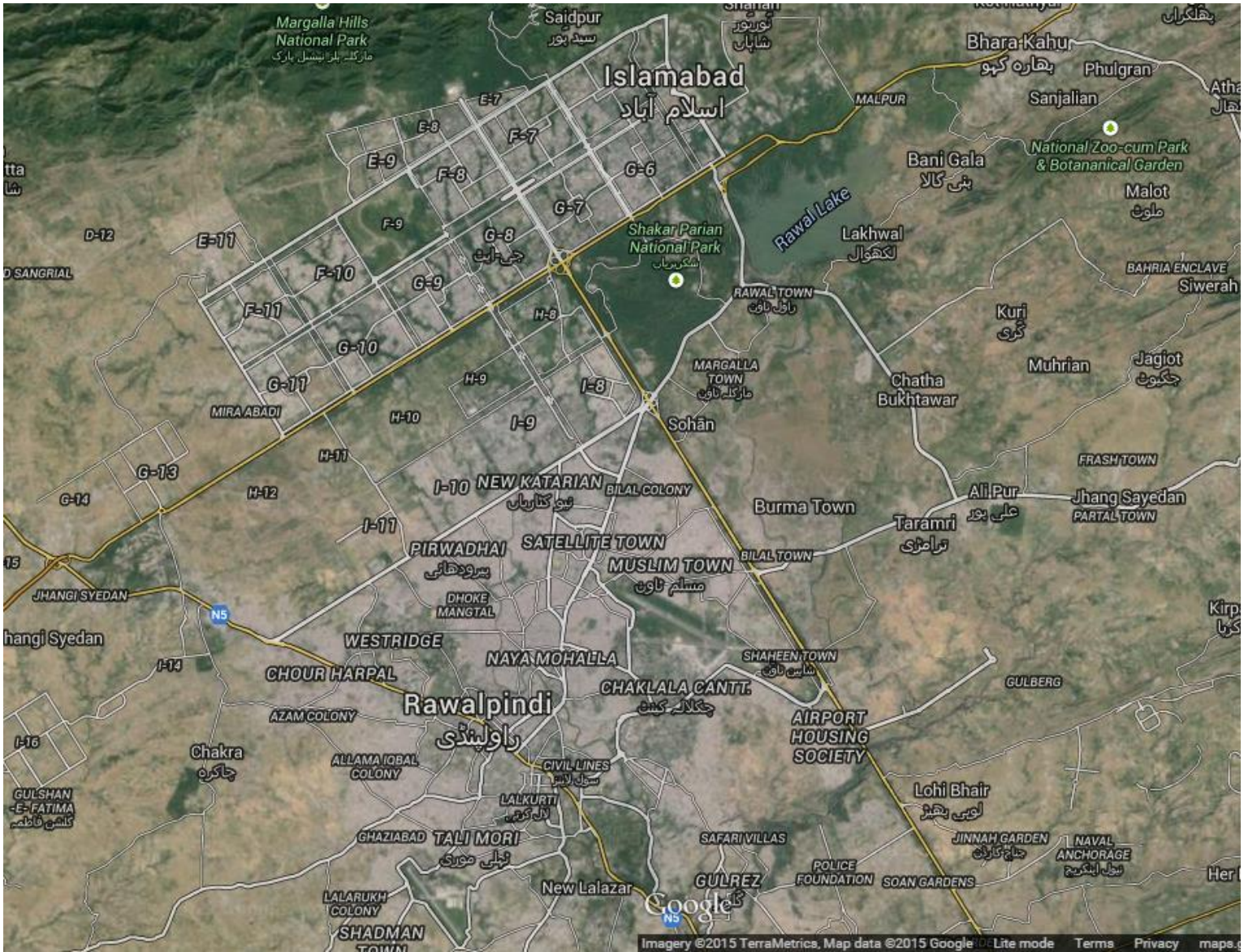


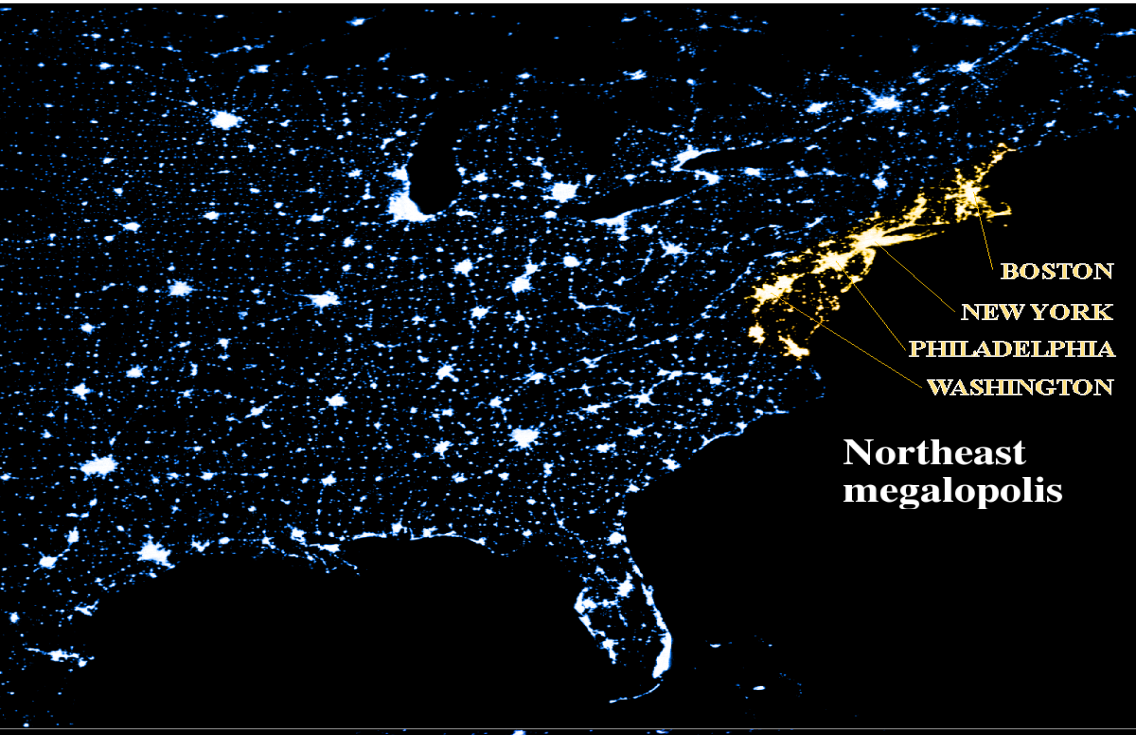
Metropolis

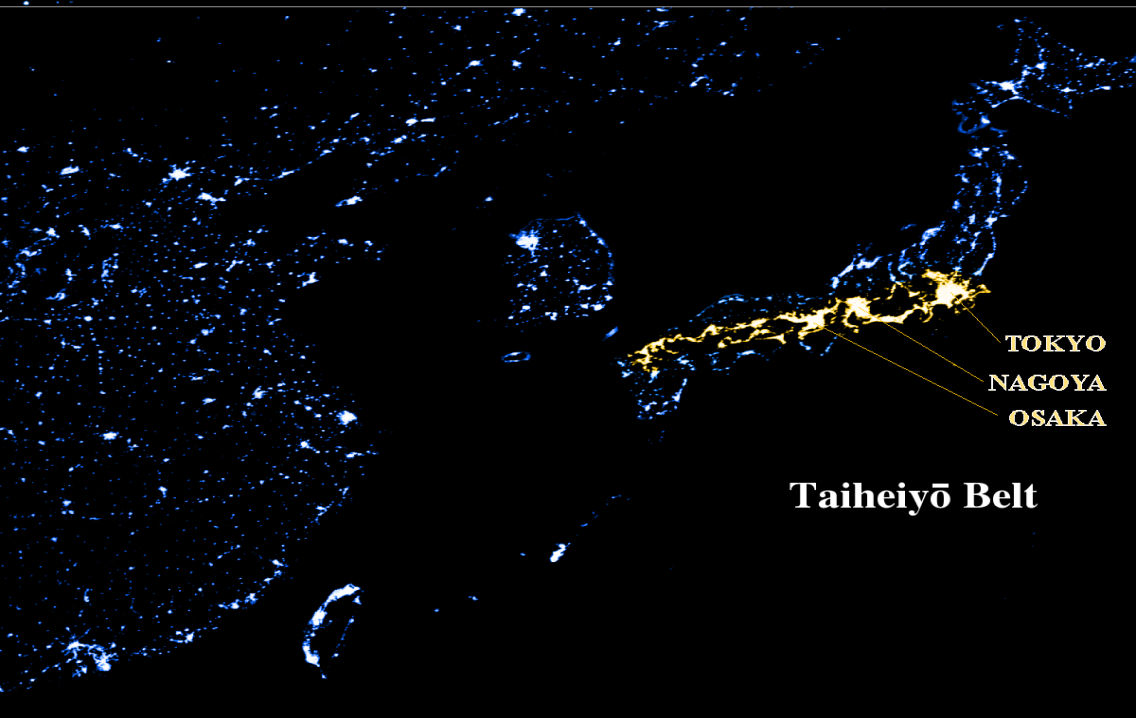


- A large city & its suburbs consisting of multiple cities & towns having 1 to 3 million people.

Conurbation/Urban Agglomeration







Megalopolis

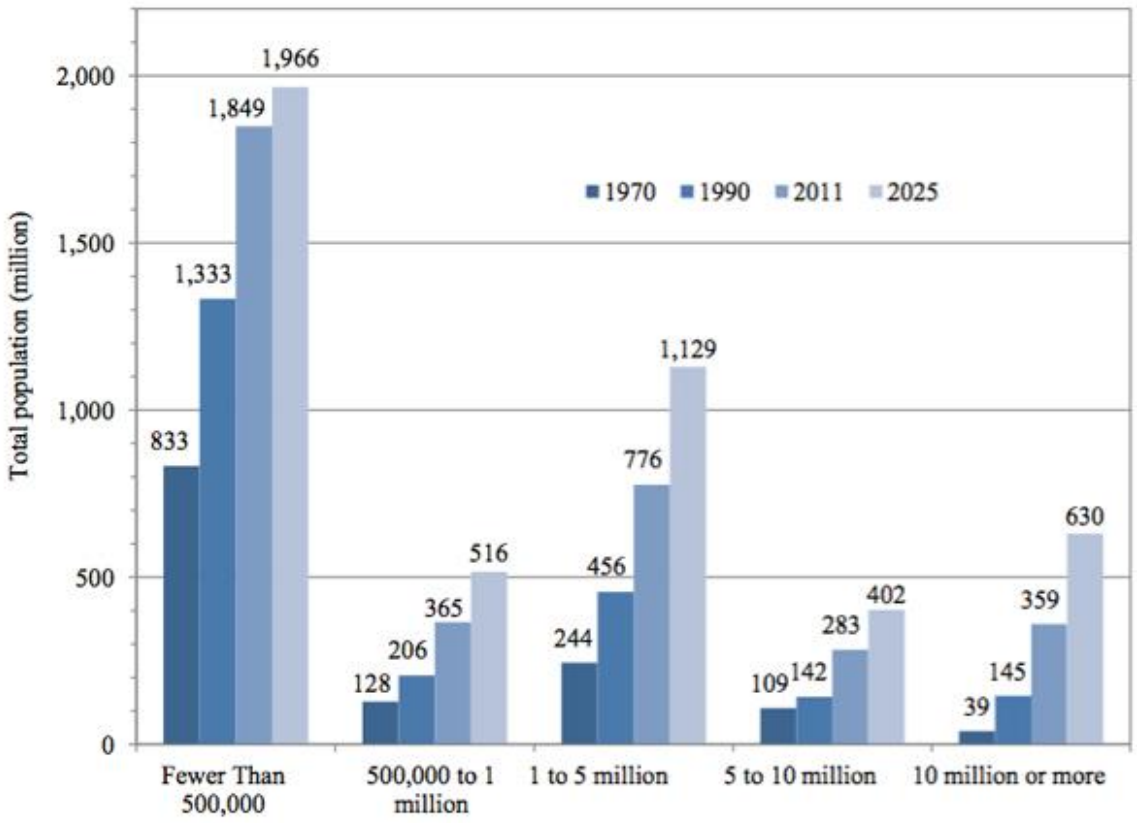
A group of conurbation, consisting of more than 10 million people.

Need of Urban Planning

Need of Town Planning (1)

- **Speed of urbanization** *and size of cities.*

Figure II. Total population in millions by city size class, 1970, 1990, 2011 and 2025

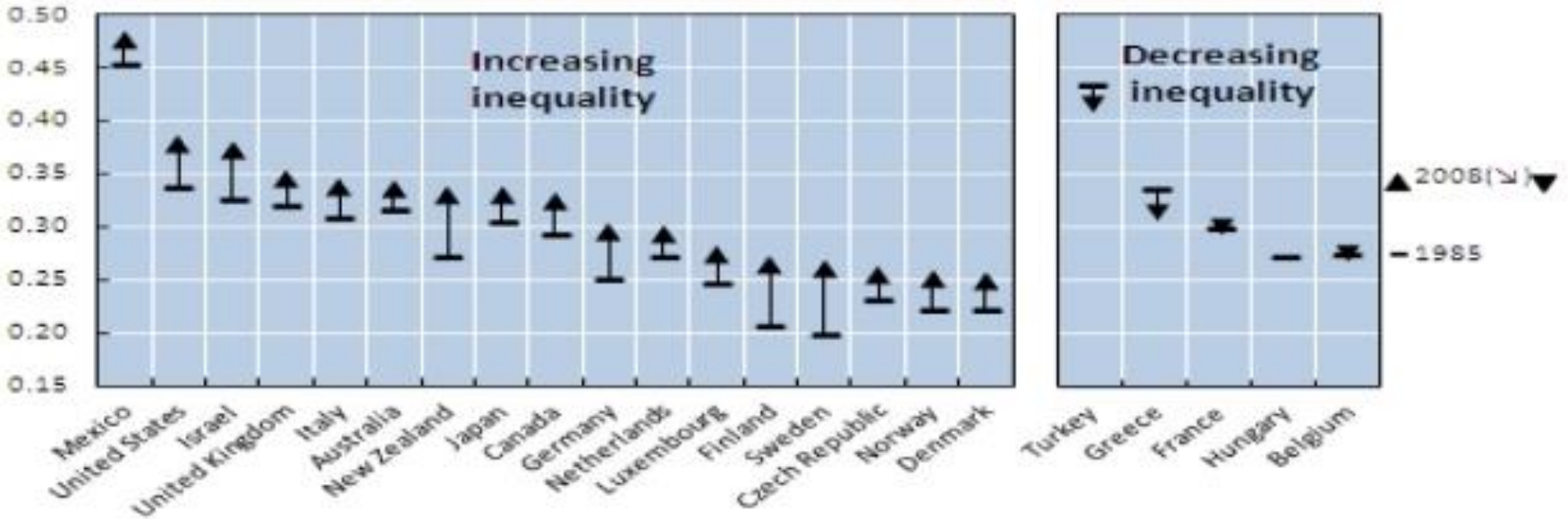


Need of Town Planning (2)

- **Social inequality**. The large number of urban poor requires special attention.

Figure 1. Income inequality increased in most OECD countries

Gini coefficients of income inequality, mid-1980s and late 2000s



Need of Town Planning (3)

Realizing cities' economic potential by overcoming weaknesses:

i) lack of vision,

ii) inadequate consultation among stakeholders

iii) un-sustainable development

iv) Lack coordination

Problems associated with Unplanned Urbanization

1. High capital and operating costs related to the provision of municipal services, with the customary tax revenues being inadequate to meet rising costs of providing municipal services to widely dispersed, rapidly expanding populations
2. Unstable tax bases for both older central city and first-ring suburban areas and newer outer-ring suburban and rural-urban fringe areas
3. Pollution of streams, lakes, and water courses
4. Drainage and flooding problems
5. Air pollution
6. Water supply problems

Problems associated with Unplanned Urbanization

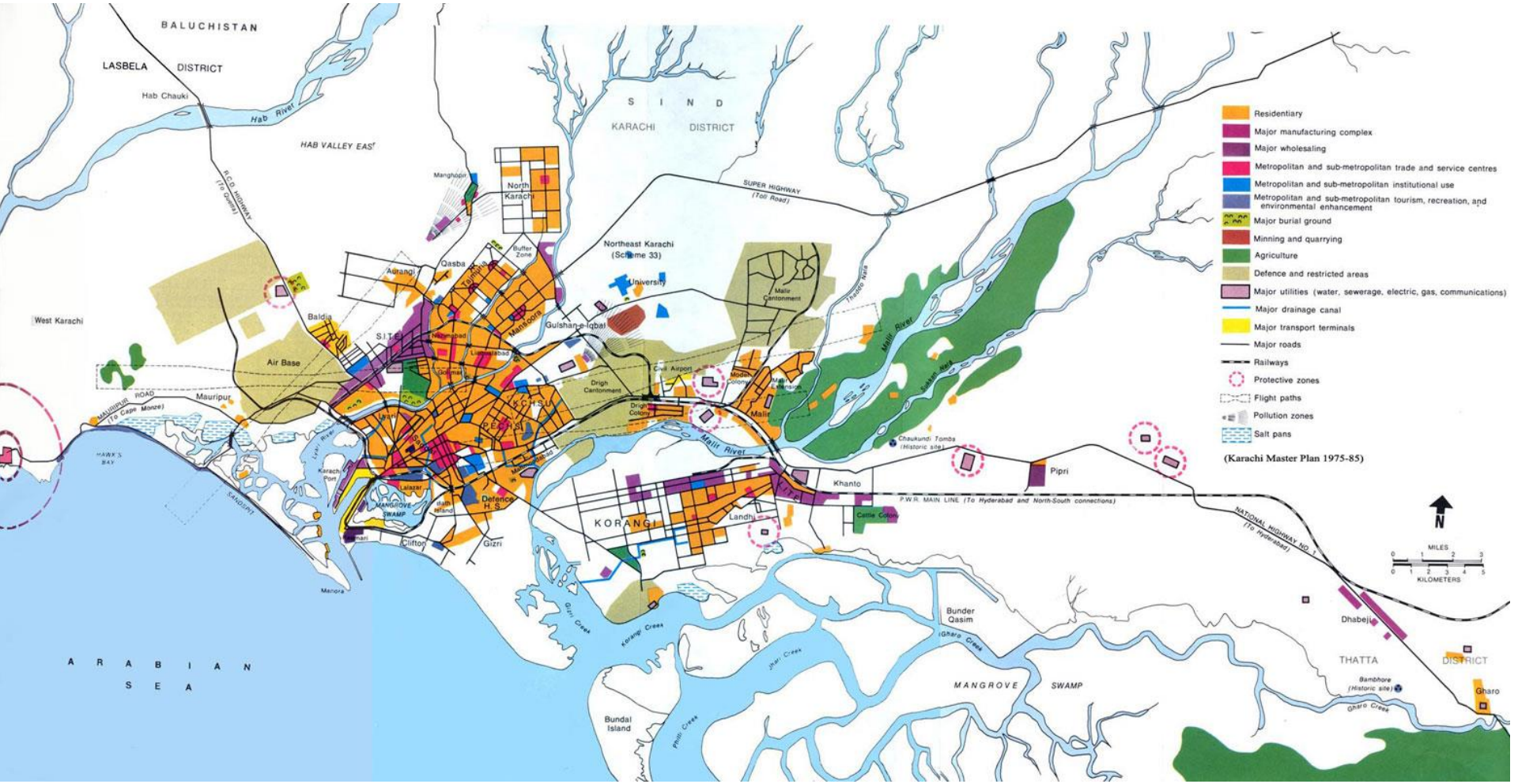
7. Power failures
8. Overcrowded and inadequate transportation facilities and community facilities, such as schools
9. Inadequate park and open space areas
10. Often irreparable misuse of the land
11. Exacerbation of the already serious problems of the central business district and surrounding core areas of the older central cities together with the appearance of blight in the older first ring suburbs
12. Substandard areas of housing lacking essential public facilities and services, and inadequate housing for the needed labor force in proximity to employment concentrations

Outcomes of Planning

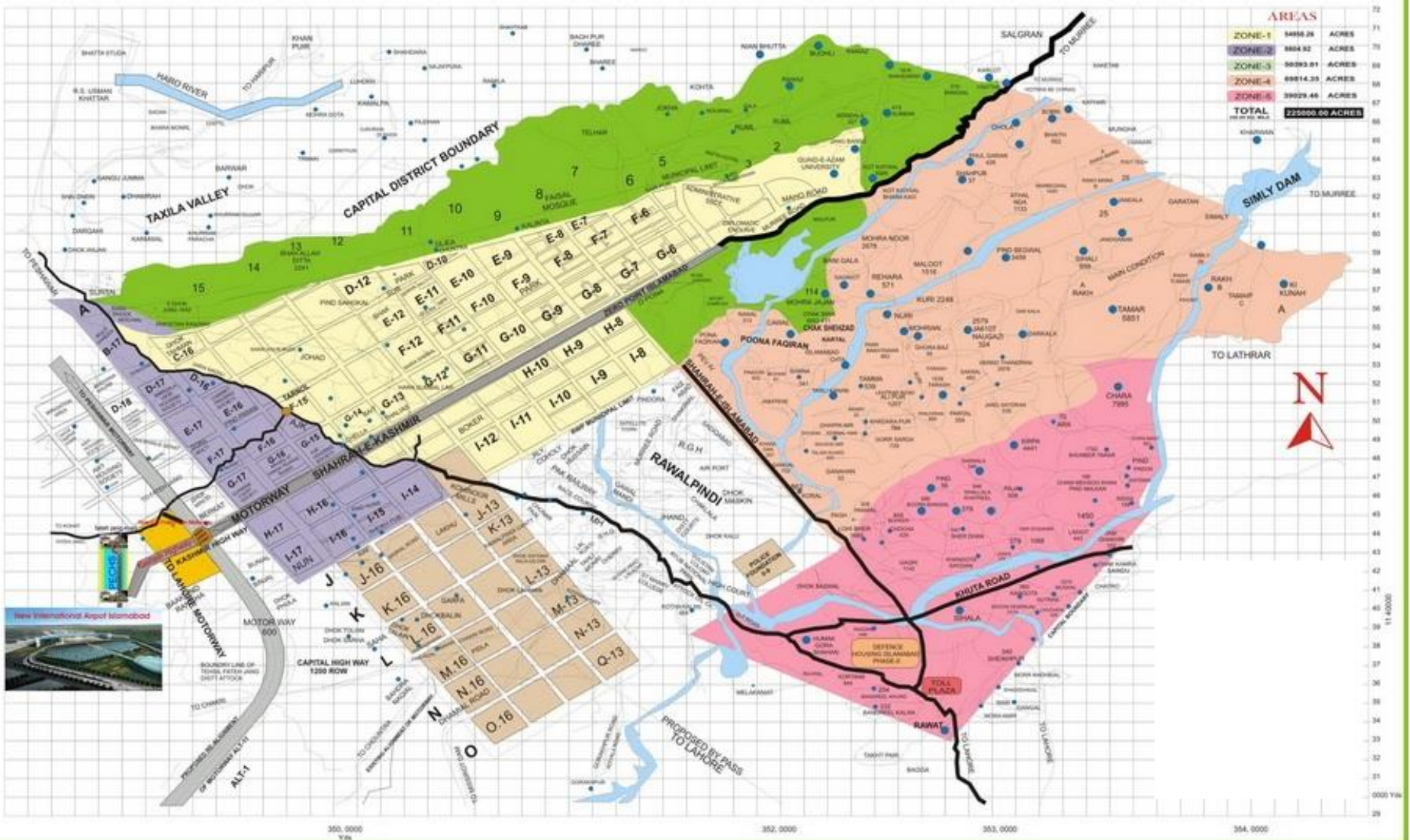
- A **plan** is typically any diagram or list of steps with timing and resources, used to achieve an objective.
- The plan is like a **Compass.**
  - It is a navigational tool which aids in following the course of development.
- Common form is MASTER PLAN, LOCAL DEVELOPMENT PLANS, STRUCTURE PLAN etc.


PROPOSED LAND USE
SINGLE / TWO FAMILY
TOWNHOUSE
MIXED USE: RESIDENTIAL + RETAIL/OFFICE
MIXED USE: LIVE-WORK
MULTI FAMILY
OFFICE
RETAIL
COMMERCIAL SERVICES
LIGHT INDUSTRY
HEAVY INDUSTRY
INSTITUTIONAL
RECREATION/OPEN SPACE/LANDSCAPE
TRANSPORTATION / PUBLIC UTILITIES
PARKING (DOWNTOWN ONLY)

Karachi


BALUCHISTAN
LASBELA DISTRICT
Hab Chauki
Hab River
HAB VALLEY EAST
S I N D
KARACHI DISTRICT
SUPER HIGHWAY (Toll Road)
North Karachi
Northeast Karachi (Scheme 33)
University
Malir Cantonment
Gulshan-e-Iqbal
West Karachi
Baldia
Air Base
Mauripur
Civil Airport
Drigh Cantonment
Malir
Malir River
Chaukundi Tombs (Historic site)
Pipri
Khanto
Cattle Colony
P.W.R. MAIN LINE (To Hyderabad and North-South connections)
NATIONAL HIGHWAY NO. 1 (To Hyderabad)
KORANGI
Landhi
Clifton
Gizri
Defence H.S.
Karachi Port
Lalazar
MANGROVE SWAMP
Manora
HAWK'S BAY
SANDSPIT
A R A B I A N S E A
Bundal Island
Bunder Qasim
Dhabeji
THATTA DISTRICT
Bambhore (Historic site)
Gharo
MANGROVE SWAMP
Residentiary
Major manufacturing complex
Major wholesaling
Metropolitan and sub-metropolitan trade and service centres
Metropolitan and sub-metropolitan institutional use
Metropolitan and sub-metropolitan tourism, recreation, and environmental enhancement
Major burial ground
Minning and quarrying
Agriculture
Defence and restricted areas
Major utilities (water, sewerage, electric, gas, communications)
Major drainage canal
Major transport terminals
Major roads
Railways
Protective zones
Flight paths
Pollution zones
Salt pans
(Karachi Master Plan 1975-85)
N
MILES
KILOMETERS

MASTER PLAN OF ISLAMABAD


AREAS
ZONE-1 54858.26 ACRES
ZONE-2 5804.92 ACRES
ZONE-3 50393.01 ACRES
ZONE-4 69814.35 ACRES
ZONE-5 39029.46 ACRES
TOTAL 225000.00 ACRES
CAPITAL DISTRICT BOUNDARY
TAXILA VALLEY
HARO RIVER
SALGRAN
TO MURREE
SIMLY DAM
TO LATHRAR
MARGALLA ROAD
SHAHRAH-E-KASHMIR
MOTORWAY
RAWALPINDI
KHAYABAN-E-JEHLAMABAD
KHAYABAN-E-SIRSYED
KHAYABAN-E-SURSYED
KHAYABAN-E-IQBAL
KHAYABAN-E-SULEMAN
KHAYABAN-E-JINNAH
KHAYABAN-E-ZAFAR
KHAYABAN-E-ATTA TURK
KHAYABAN-E-KARAKORAM
KHAYABAN-E-SARSABZ
KHAYABAN-E-MILLAT
KHAYABAN-E-GHAZNAVI
KHAYABAN-E-SHAMS
KHAYABAN-E-BADR
KHAYABAN-E-TARIQ
KHAYABAN-E-SHAHEEN
KHAYABAN-E-ROOMI
KHAYABAN-E-AZMAT
KHAYABAN-E-AZAM
KHAYABAN-E-ALI
KHAYABAN-E-SALAR
KHAYABAN-E-SHAHEED
KHAYABAN-E-SHAHEED
KHAYABAN-E-SHAHEED

Level of Town Planning

- **Local/district level Planning:** It is largely influenced by the economic conditions for the development of the town (short – medium term).
- **Regional Level Planning:** is the process of formulating and clarifying responses to certain problems with a regional dimensions (medium term).
- **National level Planning** is usually economic in nature and long term.

Goals for Town Planning

- Reduce poverty and improve quality of life for the most people.
- Distribute both positive and negative aspects of development fairly.
- Environmental conservation
- Cultural and historic preservation
- Provide a predictable process for decision making that is informed by these goals.

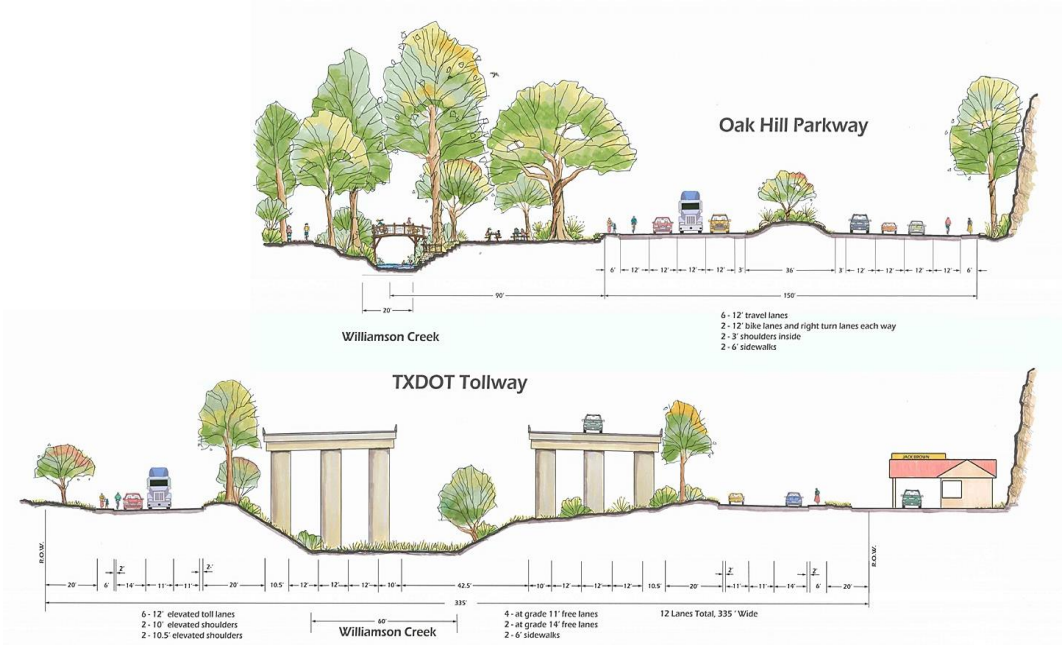
Aims and Objectives of Urban Planning

Aims and Objectives of Planning

Beauty

Beauty means to preserve the aesthetic in the design of all elements of the town or city plan

Beauty is achieved by taking the most possible advantages of the natural conditions.



Aims and Objectives of Planning

Convenience

Heavy traffic congestion during the working hours of the day





Electricity service's

Aims and Objectives of Planning

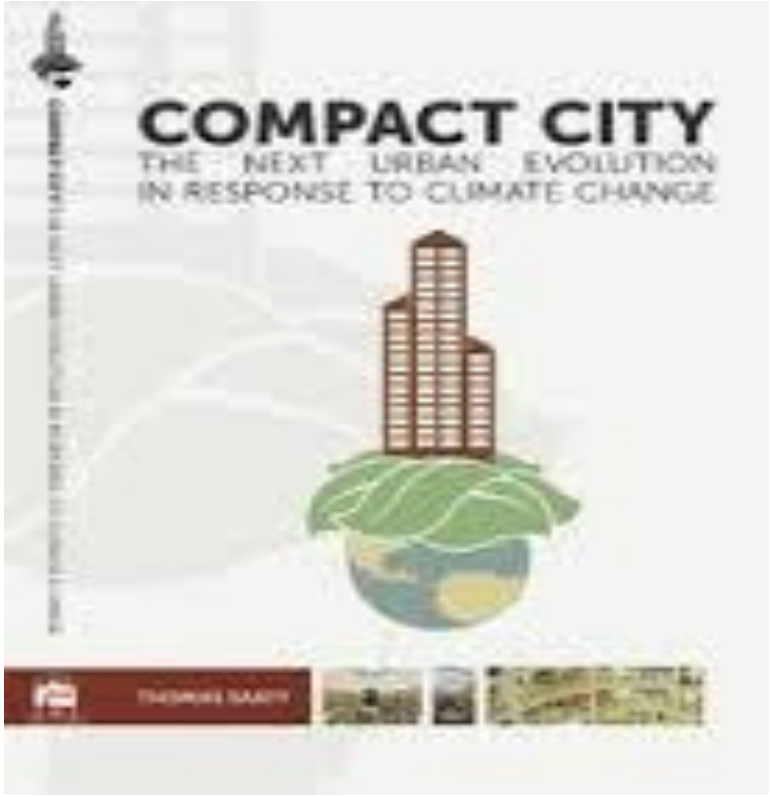
Health

- Health is accommodated in a town planning by making right use of land for purpose by providing parks for public, by maintaining the pollution of various forms.
- For providing good air to the people the inadequate open spaces for park and playgrounds resulting in unhealthy living condition
- It is governed by zoning rules.

Aims and Objectives of Planning

Convenience

Convenience is understood in the form of various economic, social and recreational amenities to be given to the public.

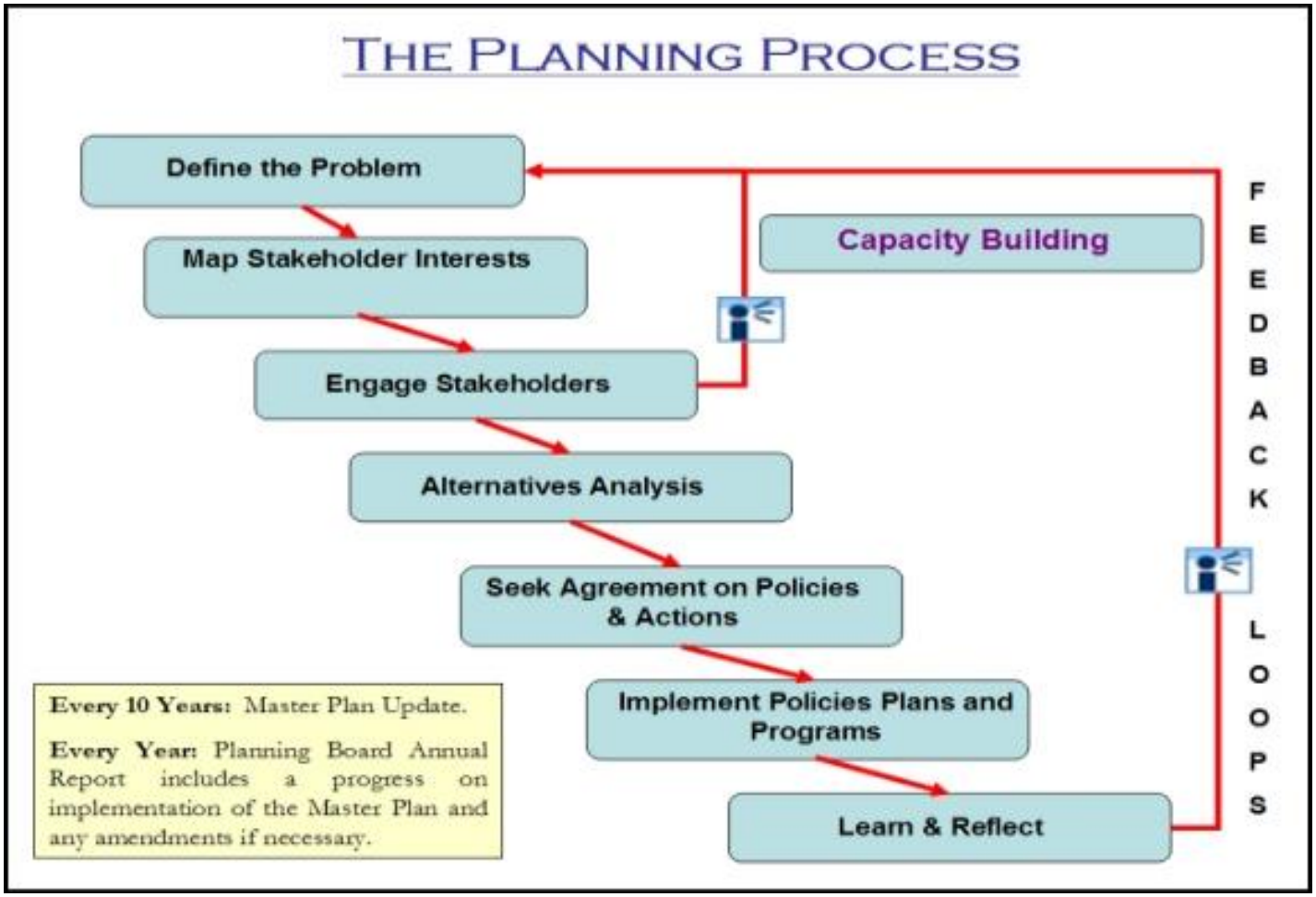




Principles of Urban Planning

Principles of Town Planning

1. No haphazard method of planning



Principles of Town Planning

2. Adequate housing for all segments of society



Pakistan is facing an unprecedented **shortage** of 7.6 million **housing** units reveals a latest World Bank report, 2009-10

Principles of Town Planning

3. Provision of civic amenities like water supply

4. Adequate open spaces



- less than 30 % Pakistani have access to safe dinking water



Principles of Town Planning

5. Accessibility….

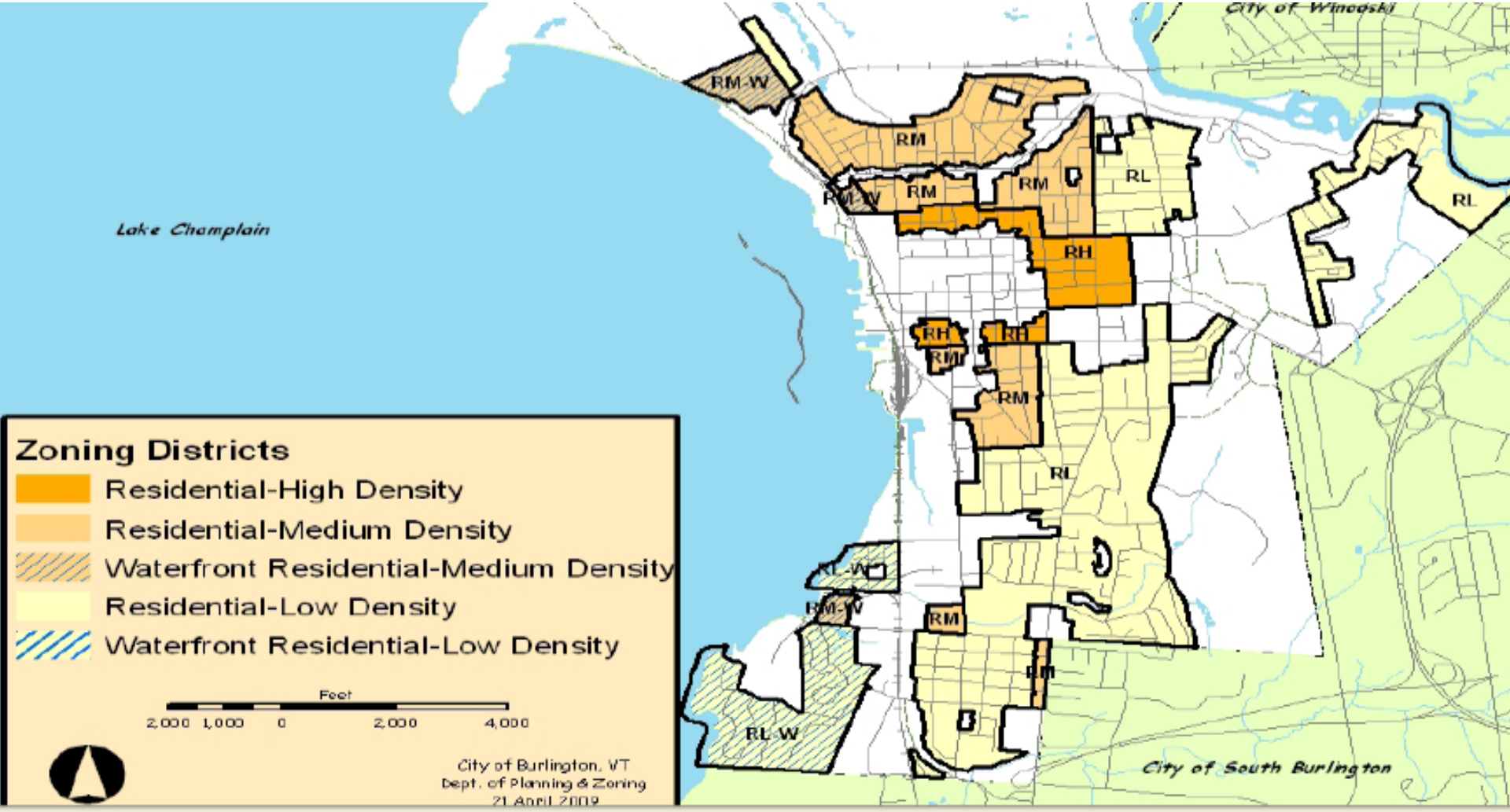


Mode of Accessibility

Rail
Road
Air
Water

Principles of Town Planning

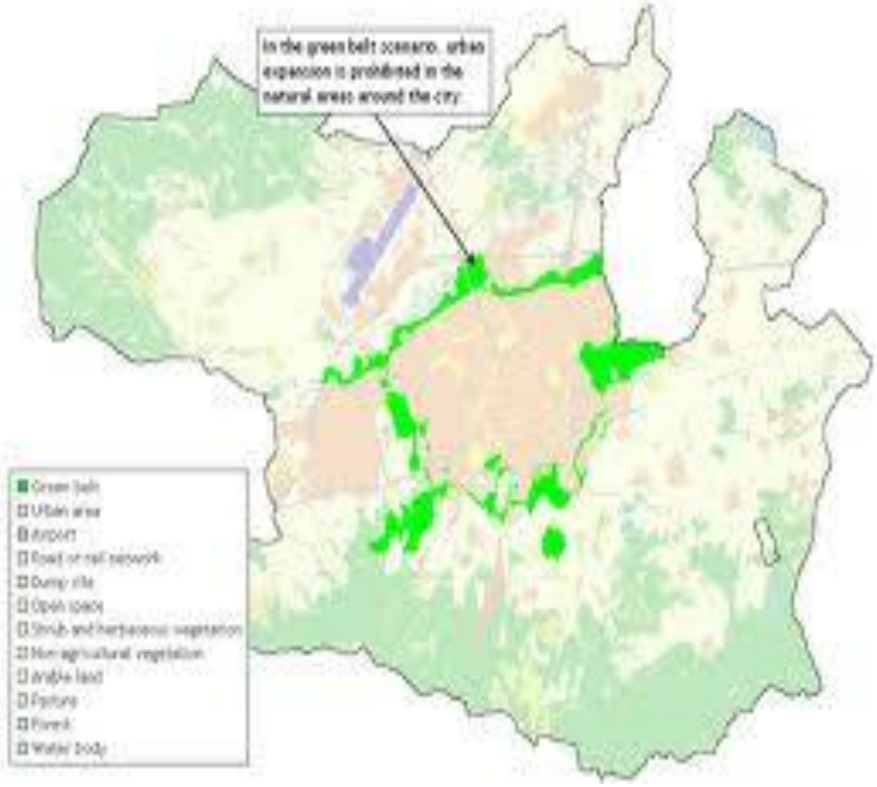
6. Systems of zoning

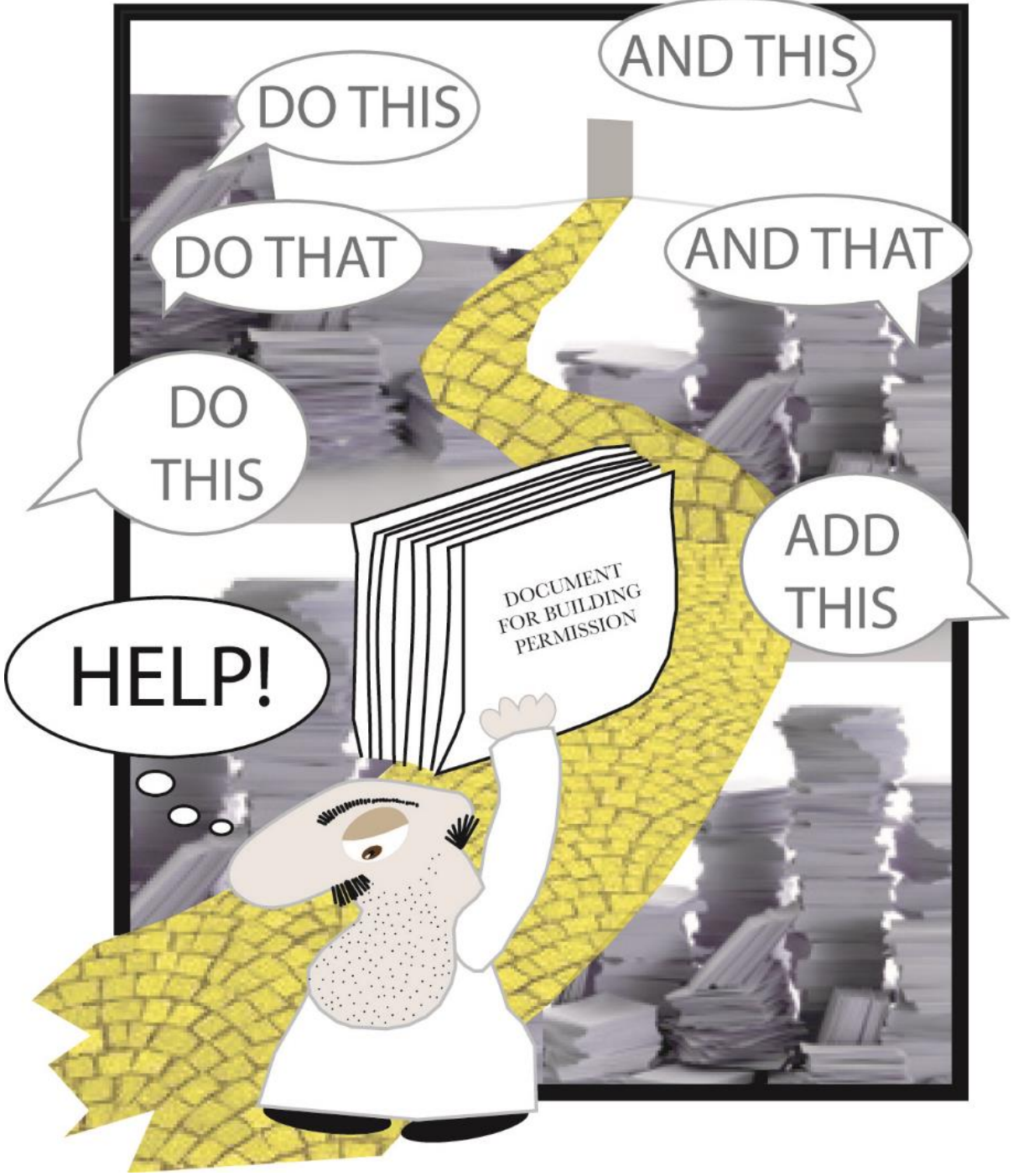


Principles of Town Planning

7. Provision of green belts to control the future growth of a city






DO THIS
AND THIS
DO THAT
AND THAT
DO THIS
ADD THIS
DOCUMENT FOR BUILDING PERMISSION
HELP!

What Planners Do?

- Planning is vital to the sustainability of safe, healthy, and secure urban environments
- A Planners deal with:
  - Land Use
  - Addressing transportation and infrastructure
  - Planning social and community services
  - Managing cultural and heritage resources
  - Creating economic capacity in local communities
  - Working internationally

Urban Planning Specializations

- Land use and Development Control
- Smart City and Technology
- Transportation Planning
- Disaster and Climate Resilient Development
- Environmental and Natural Resources Development
- Community and Housing Development
- Regional Economic Development
- Historic and Cultural Preservation
- Urban Design

Planning Specializations - American Planning Association


THANK
YOU!